Lab 6: Code and Branch Coverage Analysis

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Module under test	mobile_data_plan.py
Repository	L6_L7_45_mobile_data_plan

1. Objective

To measure and improve test coverage of **mobile_data_plan.py** using Python's *coverage* and *pytest*, reaching at least 90% line coverage and 90% branch coverage, and to understand the difference between the two.

2. Module Overview

The module has three functions: **overage_charge** (data overage by plan, roaming and loyalty; raises ValueError for an unknown plan or negative usage), **recharge_allowed** (limits depend on KYC status, amount and pending recharges) and **validity_status** (returns active, grace or expired based on days left, prepaid/postpaid and add-on). The module has 37 statements and 26 branches.

3. Code (Line) Coverage

Test file: **CodeCoverageTest_mobile_data_plan.py** (17 tests). Each function is exercised along its main paths, including both error cases, the loyalty bonus, roaming, the heavy-usage surcharge, KYC limits and all three validity states.

```
coverage run -m pytest CodeCoverageTest_mobile_data_plan.py
coverage report -m
```

```
anshvardhansingh@Anshs-MacBook-Air Lab 6 % coverage erase
coverage run -m pytest CodeCoverageTest_mobile_data_plan.py
coverage report -m
========================================= test session starts ==========================================
platform darwin -- Python 3.12.5, pytest-9.2.0, pluggy-1.7.0
rootdir: /Users/anshvardhansingh/Desktop/L6_L7_45_mobile_data_plan/Lab 6
plugins: hydra-core-1.3.7, anyio-4.13.0
collected 17 items

CodeCoverageTest_mobile_data_plan.py .................                                        [100%]

=========================================== warnings summary ===========================================
../../../../../Library/Frameworks/Python.framework/Versions/3.12/lib/python3.12/site-packages/an
yio/pytest_plugin.py:14
  /Library/Frameworks/Python.framework/Versions/3.12/lib/python3.12/site-packages/anyio/pytest_p
lugin.py:14: PytestRemovedIn10Warning: _pytest.python.CallSpec2 has been renamed to CallSpec.
  The CallSpec2 alias will be removed in pytest 10.
  Update imports to use CallSpec instead.
  See https://docs.pytest.org/en/stable/deprecations.html#callspec2-renamed
    from _pytest.python import CallSpec2

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
================================== 17 passed, 1 warning in 0.03s ===================================
Name                                  Stmts   Miss  Cover   Missing
-------------------------------------------------------------------
CodeCoverageTest_mobile_data_plan.py      38      0   100%
mobile_data_plan.py                       37      0   100%
```

Figure 1: Line coverage terminal output - 17 passed, mobile_data_plan.py at 100%.

4. Branch Coverage

Test file: **BranchCoverageTest_mobile_data_plan.py** (32 tests). Every decision is driven to both True and False, including boundary values: data_gb = 0, usage exactly equal to the included GB, extra = 10 vs 11, recharge amount 200/201 and 2000/2001, pending recharges 2/3, and day cutoffs for grace/expired. The loyal premium case (no bonus) and every prepaid, postpaid and add-on combination are also covered.

```
coverage run --branch -m pytest BranchCoverageTest_mobile_data_plan.py
coverage report -m
```

```
BranchCoverageTest_mobile_data_plan.py ..............................                [100%]

========================================= warnings summary =========================================
../../../../../Library/Frameworks/Python.framework/Versions/3.12/lib/python3.12/site-packages/an
yio/pytest_plugin.py:14
  /Library/Frameworks/Python.framework/Versions/3.12/lib/python3.12/site-packages/anyio/pytest_p
lugin.py:14: PytestRemovedIn10Warning: _pytest.python.CallSpec2 has been renamed to CallSpec.
  The CallSpec2 alias will be removed in pytest 10.
  Update imports to use CallSpec instead.
  See https://docs.pytest.org/en/stable/deprecations.html#callspec2-renamed
    from _pytest.python import CallSpec2

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
================================ 32 passed, 1 warning in 0.05s =================================
Name                                    Stmts   Miss Branch BrPart  Cover   Missing
-----------------------------------------------------------------------------------------
BranchCoverageTest_mobile_data_plan.py      68      0      0      0   100%
mobile_data_plan.py                         37      0     26      0   100%
-----------------------------------------------------------------------------------------
TOTAL                                      105      0     26      0   100%
```

Figure 2: Branch coverage terminal output - 32 passed, mobile_data_plan.py at 100% (26/26 branches).

5. Results Summary

Type	Test file	Tests	Result on mobile_data_plan.py
Line coverage	CodeCoverageTest_...py	17	100% (37/37 statements)
Branch coverage	BranchCoverageTest_...py	32	100% (26/26 branches)

6. Line vs Branch Coverage

Line coverage only checks that each line ran at least once. Branch coverage checks that each decision went both ways. For example, one test with a roaming user and heavy usage runs the surcharge lines, but only tests with extra = 10 or a non-roaming user show that the 'extra > 10' and 'is_roaming' decisions also behave correctly when False. The branch tests therefore catch cases the line tests alone can miss.

7. Conclusion

Both test suites exceed the 90% requirement, reaching 100% line and 100% branch coverage with all tests passing. The test files, screenshots and this report were committed to GitHub.